

30th Annual CASIS Workshop 2026

Day 1 - Wednesday, June 24



ENGINEERING
LAWRENCE LIVERMORE NATIONAL LABORATORY



SAN FRANCISCO | OAKLAND EAST BAY



**UNIVERSITY
OF
CALIFORNIA**
Livermore
Collaboration
Center

Time		Program
Start	End	
7:30	17:30	Registration
8:00	8:10	Welcome CASIS Director Ruben Glatt (LLNL)
Non-Destructive Evaluation (Seemeen Karimi, Harry Martz)		
8:10	8:30	Science and Technology on a Mission: An Overview of LLNL and The Nondestructive Characterization Institute Featured Speaker: Harry Martz (LLNL)
8:30	8:40	High-Energy Tomographic Scanner for Inspection of Contraband in Air Cargo Joseph Bendahan (LLNL)
8:40	8:50	Lab to Fab: High-Resolution 3D X-Ray at Production Speeds Jeff Gelb (Applied Materials)
8:50	9:00	Using resonant ultrasound spectroscopy to characterize defects in manufactured metal parts Jordan Lum (LLNL)
9:00	9:10	Distributed Stochastic Optimization of a Neural Representation Network for Time-Space Tomography Reconstruction K. Aditya Mohan (LLNL)
9:10	9:20	Critical Mineral Assay at 300 Tons per Day: An NDE Gap in Secondary Feedstocks Jorge Osio-Norgaard (PHNX Materials, Inc)
9:20	9:30	Compact Multi-GSa/s Waveform-Digitizing ASICs for Single-Shot Diagnostics and High-Channel-Count Detector Readout Benjamin Rotter (Nalu Scientific)
9:30	9:40	X-ray Computed Tomography and Machine Learning John C. Miers (Sandia National Laboratories)
9:40	10:00	Automating X-ray Microscopy for Self Driving Labs with Agentic Controllers Featured Speaker: Nathan Johnson (CARL ZEISS RESEARCH MICROSCOPY SOLUTIONS)
10:00	10:30	Coffee Break + Poster Session
AI / Machine Learning (Phan Nguyen, Kowshik Thopalli)		
10:30	10:50	AI at LLNL Featured Speaker: Peer-Timo Bremer (LLNL)
10:50	11:00	Learning Nuclear Cross Sections Across the Chart of Nuclides with Local-Context Transformers Hongjun Choi (LLNL)
11:00	11:10	Edge Machine Learning for Smart Detectors with Embedded FPGAs Julia Gonski (SLAC) Amar Saini (LLNL)
11:10	11:20	LanceDB for High-Performance Image Search: A Multimodal Vector Database Approach Arun Kumar Mathiyazhagan (Amazon)
11:20	11:30	Protocol-Level Efficiency in Agent Communication: An Empirical Comparison of NLP and A2A Ranjan Sinha (IBM)
11:30	11:40	VIPER: A Video Intelligence and Perception Engine for Reasoning Amar Saini (LLNL)
11:40	11:50	Tensor Methods: A Unified and Interpretable Approach for Material Design Shaan Pakala (University of California Riverside)
11:50	12:00	Machine Learning and Artificial Intelligence applications in the HPC4EI Program Vic Castillo (LLNL)
12:00	12:50	Lunch Break sponsored by IEEE Signal Processing Society + Poster Session
Robotics & Automation (Aldair Gongora, Abhik Sarkar)		
12:50	13:10	Robotics and Automation at LLNL Featured Speaker: Aldair Ernesto Gongora (LLNL)
13:10	13:20	Automated Photochemical Screening Using an Intelligent Robotic Platform Sarah Finnegan (University of Wisconsin Madison)
13:20	13:30	Robotic Automation for Polymer Innovation and Discovery (RAPID) Michael Ford (LLNL)
13:30	13:40	Lab Automation and Microfluidics for Machine Learning-Driven Biomanufacturing Kshitiz Gupta (LLNL)
13:40	13:50	Designing and Applying a Robotic Platform for Automated Polymer Aging (PolyAge) Karly Knox (LLNL)
13:50	14:00	Integrating Robotics and Automation into High-Throughput Polymer Science Workflows Rob Learsch (LLNL)
14:00	14:10	Studying-Polymers-On a-Chip: Automated, high-throughput screening of polymer membranes for energy applications Johanna Schwartz (LLNL)
14:10	14:20	Automated Screening to Build Metalloprotein Datasets for Critical Metal Recovery Patrick Diep (LLNL)
14:20	14:30	Global Security Implications of Neuromorphic Imagers for Robotics and Automation David Mascarenas (Los Alamos National Laboratory)
14:30	15:00	Coffee Break + Poster Session
Remote Sensing, Non-Invasive Imaging & Inverse Problems (Sean Lehman, Viacheslav Li)		
15:00	15:20	Plenoptic Radiation Imager for Spectroscopic and Directional Mapping Featured speaker: Xiang Dai (University of California San Diego)
15:20	15:30	Localized Electrochemical Metal 3D Printing Abraham Akinin (LLNL)
15:30	15:40	Learning Forward and Inverse Operators for Complex Atmospheric Trace Gas Emission Source Localization Derek Hollenbeck (University of California Merced)
15:40	15:50	Superresolution Imaging of the Urban Subsurface Environment Using Telecommunications Fiber Optic Sensors Eric Matzel (LLNL)
15:50	16:00	Operational Characterization of LAPPD Gen 2: From Readout to Simulation Shang-Wen Stradleigh (University of Rochester)
16:00	16:10	Tomographic Volumetric AM with Coupled Differentiable Ray-Optical and Photochemical Optimization using Dr.TVAM Felix Wechsler (EPFL)
16:10	16:30	Obsessive Second Order System Study: A simple question's answer went rogue Featured speaker: Sean K. Lehman (LLNL)
30 years CASIS workshop (Dave Chambers)		
16:30	16:45	Leadership Address Kim Budil, Anup Singh, Rob Sharpe
16:45	17:30	CASIS through the years Dave Chambers (LLNL)
16:45	17:30	CASIS Panel (Noah Pfuegler) Dave Chambers, Jim Candy, Randy Roberts, Steve Azevedo
17:30	19:00	Happy Hour



30th Annual CASIS Workshop 2026		
Day 2 - Thursday, June 25		
Time	Start	End
7:30	17:00	Registration
8:00	8:10	Welcome CASIS Director Ruben Glatt (LLNL)
Robotics & Automation (Aldair Gongora, Abhik Sarkar)		
8:10	8:30	VLMs, Scene Understanding, and Robot Learning Featured Speaker: Amit Roy Chowdhury (University of California Riverside)
8:30	8:40	From Language to Action: Orchestrating Robots and Instruments with Off-the-Shelf LLMs Henry Williams (LLNL)
8:40	8:50	Dual-Sensing Strategies for Pneumatic Soft Grippers in Delicate Manipulation Tasks Teodoro Lima (University of California Merced)
8:50	9:00	Autonomous Defect Detection and Process Optimization in Binder Jet Printing Brennan Birn (University of California Berkeley)
9:00	9:10	Computer Vision-Guided Robotic Fill-Tube Alignment and Assembly for Fully 3D Printed NIF Capsules Sydney Gothenquist (LLNL)
9:10	9:20	Multimodal LLMs as Oversight Tools in Robotic Laboratories Shahryar Mooraj (LLNL)
9:20	9:30	Deployable and Formally Verified Robotic Mission Planning using Natural Language Driven Behavior Trees Marcos Zuzuarregui (University of California Merced)
9:30	9:40	Physics-informed in-silico optimization for structural design and 3D printing of electromag. interference shielding materials Christopher Spezzano (University of California Berkeley)
9:40	9:50	MatGym: A Framework for Human-AI Co-Design in Architected Materials Research Rodrigo Telles (LLNL)
9:50	10:00	Industrial Automation Enabled Self-Driving Workflow for Rapid Materials Exploration Mason Sage (LLNL)
10:00	10:30	Coffee Break + Poster Session
National Ignition Facility (Eugene Kur, Christopher Miller)		
10:30	10:50	Electromagnetic Interference Testing and Protection of Electronics for Z-pinch X-ray Diagnostics Featured Speaker: Anne Garafalo (Pacific Fusion)
10:50	11:00	Memory-Augmented Few-Shot Segmentation Mask Generation via SNAP: Smart Neural Annotation Pipeline Md Khayrul Islam (LLNL)
11:00	11:10	Toward ML-Driven ICF Experiment Design at NIF with Exascale Optimization on El Capitan Bogdan Kustowski (LLNL)
11:10	11:20	From Scripts To AI Agents, Scaling Scientific Campaigns M. Giselle Fernandez-Godino (LLNL)
11:20	11:30	Closed-Loop Autonomy and Workflow Integration in High-Repetition-Rate HED Experiments Abhik Sarka (LLNL)
11:30	11:40	Igniting the Future through High-Precision, High-Throughput Additive Manufacturing Dongping Terrel-Perez (LLNL)
11:40	12:00	Plasma probing and single-shot crossed-beam energy transfer measurements using a broadband probe Featured Speaker: Andrew Longman (LLNL)
12:00	13:00	Lunch Break sponsored by IEEE Computer Society + Poster Session
Quantum Sensing & Quantum Computing (Kristi Beck)		
13:00	13:20	Quantum Sciences at LLNL Featured Speaker: Kristin Beck (LLNL)
13:20	13:30	Space Ghost: Quantum Imaging from Orbit Shervin Kiannejad (LLNL)
13:30	13:40	STEM Characterization of Alternating Bias Assisted Annealing Influences on the Structure of Amorphous Alumina in Superconducting Quantum Resonators Nicholas Hagopian (University of Wisconsin Madison)
13:40	13:50	Non-equilibrium Dynamics of Two-level Systems directly after Cryogenic Alternating Bias Vito Iaia (LLNL)
13:50	14:00	Developing and Characterizing the 3D Quantum Microscope Audrey A. Eshun, Ted Laurence (LLNL)
14:00	14:10	Non-equilibrium energy flow and correlated error mechanisms in quantum and cryogenic devices Sergey Pereverzev (LLNL)
14:10	14:20	Second quantization model of the Vlasov-Poisson system for quantum computation Michael May (LLNL)
14:20	14:30	Quantum Entanglement and Magic in Nuclear Processes Saurabh V. Kadam (LLNL)
14:30	14:40	Adding Gravity Gradiometry to Other Sensor Modalities Sean Walston (LLNL)
14:40	15:00	New developments in Quantum Sensing with Electron and Nuclear Spins Featured Speaker: Ashok Ajoy (University of California Berkeley)
15:00	15:30	Coffee Break + Poster Session
AI / Machine Learning (Phan Nguyen, Kowshik Thopalli)		
15:30	15:50	Targeted Unlearning with Single Layer Unlearning Gradient Featured Speaker: Salman Asif (University of California Riverside)
15:50	16:00	From Missing Concepts to Weak Steering: Refining Sparse Autoencoders Akshay Kulkarni (University of California San Diego)
16:00	16:10	Quality Gates for AI: A Unified MLOps Framework for Trustworthy, Cloud-Native ML Systems Raja Rao Budaraju (Oracle America Inc)
16:10	16:20	Scaling inverse folding with protein-protein interaction and predicted structures Sudeep Sarma (LLNL)
16:20	16:30	HLaSDI: Reduced order modeling for higher order dynamics Robert Stephany (LLNL)
16:30	16:40	Graph Grammar Learning for Automatic Circuit Generation Qing Wang (HKU)
16:40	16:50	Task-Aware Optimization of Reasoning Data Mixtures for Large Language Model Post-Training Gautam Singh (LLNL)
16:50	17:00	Closing

